Request for Information (RFI) (#RIN 0955–AA13): Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

Office of the Deputy Secretary and Assistant Secretary for Technology Policy (ASTP) and Office of the National Coordinator for Health Information Technology (ONC), Department of Health and Human Services (DHHS)

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Submitted by:

Campbell Buhrow
Account Executive
+1 703-310-5722
campbell.buhrow@sas.com

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Executive Summary

The U.S. Department of Health and Human Services (HHS) is seeking to accelerate the responsible adoption and use of artificial intelligence (AI) in clinical care. While AI technologies have advanced rapidly, their integration into routine care delivery remains inconsistent, fragmented, and slowed by operational and governance complexity. HHS is uniquely positioned to shape a national framework that enables innovation while preserving patient safety, equity, privacy, and public trust. SAS appreciates the opportunity to respond to this Request for Information and to share our perspective on how HHS can catalyze scalable, trustworthy AI adoption across the healthcare ecosystem.

HHS is confronting a pivotal challenge: how to unlock AI's transformative potential while ensuring safety, accountability, and equitable outcomes at the enterprise and population scale. Despite significant innovation, implementation remains hindered by siloed data, limited interoperability, workforce readiness gaps, and governance structures not designed for adaptive, continuously learning systems. Legal and operational uncertainty, particularly around model accountability, bias monitoring, performance drift, and lifecycle oversight, further constrains responsible deployment. Without a coordinated national approach grounded in operational rigor, AI will remain confined to isolated pilots rather than driving durable, system-wide improvements in quality, efficiency, and patient outcomes.

Many healthcare organizations are experimenting with AI in clinical decision support, predictive risk modeling, workflow automation, and other use cases. However, these efforts often rely on fragmented tools or conceptual governance overlays rather than deeply integrated, enterprise-grade platforms. HHS's priorities like protecting patient privacy, advancing health equity, reducing clinician burden, and modernizing care delivery, require infrastructure that unifies data, AI development, deployment, monitoring, and governance within a single operational framework. SAS envisions a healthcare ecosystem in which AI systems are transparent, bias-tested, and continuously monitored. These AI systems should be explainable by design, and seamlessly embedded into clinical workflows, delivering measurable improvements in safety, quality, and cost performance at both individual and population levels.

SAS enables the healthcare ecosystem of HHS's future through a fully mature, end-to-end AI and analytics platform purpose-built for regulated environments. Unlike solutions that layer governance onto experimental AI tools, SAS embeds model management, explainability, bias detection, version control, and performance monitoring directly into the core analytics engine. Our high-performance platform integrates structured and unstructured data at scale, supports real-time and batch analytics, and operationalizes models reliably across enterprise and multi-institution environments. SAS's privacy-enhancing technologies enable collaboration and innovation while minimizing exposure of sensitive patient information. This unified approach ensures AI is not only developed responsibly, but deployed, monitored, and sustained reliably in production environments.

With decades of experience supporting federal health agencies and other highly regulated industries, SAS brings proven, high-reliability analytics capabilities that extend beyond governance principles into real-world operationalization. SAS uniquely combines advanced analytics performance, population-level modeling expertise, embedded trust mechanisms, and enterprise-scale deployment capability within a single, interoperable platform. By partnering with SAS, HHS and its stakeholders can transition from fragmented AI pilots to a secure, scalable, and accountable ecosystem. This ecosystem can be one in which clinicians experience reduced burden, administrators gain defensible insight into outcomes and ROI, regulators maintain continuous transparency and oversight, and patients benefit from safer, more equitable care powered by AI.

RFI Response

II. Solicitation of Public Comments Regulation

Regulation

As the nation's principal health regulator, HHS helps shape the environment in which AI for clinical care is developed, evaluated, and deployed. HHS seeks to establish a regulatory posture on AI that is well understood, predictable, and proportionate to any risks presented to enable rapid innovation while protecting patients and the confidentiality of their identifiable health information, and maintaining public trust. We seek feedback on how current HHS regulations impact AI adoption and use for clinical care.

SAS appreciates HHS's commitment to fostering rapid innovation, while safeguarding patient privacy and public trust. As a global leader in advanced analytics and AI for five decades, SAS believes that clear, risk-based regulatory guidance can accelerate responsible AI adoption across clinical settings. Current HHS regulations, particularly those governing patient data privacy, information sharing, and interoperability shape the environment in which AI for clinical care is developed, evaluated, and deployed. SAS has a long history supporting healthcare organizations in developing, validating, and operationalizing AI solutions in ways that strengthen transparency, governance, and patient protections. Our feedback on how current HHS regulations impact AI adoption and use for clinical care derives from the following customer success stories:

- Getting ahead of the next pandemic with AI and machine-learning fueled drug discovery | SAS
- Duke Health and SAS Formalize Strategic Analytics and AI Collaboration | Duke Health
- Research Roundup: SAS Brings Potential for New Research on the All of Us Researcher Workbench | All of Us Research Program | NIH
- CEO Roundtable on Cancer: Can data sharing lead to cancer discoveries?
- DNA analysis unlocks better health for communities | SAS

Across healthcare organizations, AI adoption is most often slowed not by lack of interest or value, but by uncertainty around how to implement AI safety and compliantly. Current regulatory requirements and guidance can be difficult to interpret consistently across organizations, especially since AI systems evolve rapidly and require ongoing monitoring after deployment. Because of this, organizations and agencies may delay adoption or deploy AI in fragmented ways that increases operations risks and reduces transparency.

SAS believes HHS can meaningfully accelerate responsible AI adoption by providing clearer expectations for AI governance, validation, and monitoring (explicitly linked to how current regulations apply to AI) while maintaining flexibility for innovation. SAS outlines how a few current HHS regulations impact AI adoption below:

HIPAA Privacy and Security Rule Requirements: AI solutions that ingest, analyze, or generate output using PHI are subject to HIPAA requirements, including privacy limitations on use/disclosure and security safeguards. These guardrails are necessary for trust but can slow deployment when uncertainty around how HIPAA obligations apply to evolving and modern AI workflows, so consistent and practical implementation is necessary.

Information Blocking Regulations: The information blocking framework is essential for improving access to electronic health information, which is also essential for AI development and performance monitoring. However, organizations may be unsure about how to apply privacy and security limits while still meeting expectations for data access, especially across vendors and external systems.

Interoperability and Health IT Certification Rules: Clinical AI depends on consistent, high-quality, interoperable data, and traceable integration in clinical workflows. The HHS ONC

Health IT Certification Program requirements shape how EHR platforms expose data and how capabilities like security, auditability, access controls, and interoperability are implemented. In practice, differences in how systems are implemented and limits on pulling data and maintaining audit trails make it harder to deploy AI consistently across settings.

Clarity for Non-Medical Devices: For AI tools that fall outside traditional FDA medical device classifications, organizations often face uncertainty about where regulatory expectations begin and end, reinforcing the need for HHS-led guidance on governance, evaluation, and monitoring for non-device clinical AI.

HIPAA Right of Access and Individual Access Rule: HIPAA Right of Access requirements raise important questions about how AI-generated insights, summaries, or recommendations should be treated within the designated record set, particularly as patient-facing AI tools become more common. Better clarity on how these requirements apply to AI outputs would support transparency, patient engagement, and consistent implementation across healthcare organizations.

Opportunities to Support Responsible AI Adoption: SAS believes HHS can meaningfully accelerate responsible adoption by providing clearer expectations for AI governance, validation, and monitoring, while maintaining flexibility for innovation. SAS can support these objectives by helping to:

- Establish clear and repeatable AI governance practices that embed auditability, accountability, and oversight across model development, validation, deployment, and monitoring. Strong lifecycle practices help organizations apply HIPAA, civil rights, and other regulatory requirements consistently when using AI in clinical care.
- Promote transparency through standardized model documentations (e.g., model cards) that explain what an AI system is intended to do, its limitations, how it was trained and evaluated, how well it performs, and how it should be monitored. Clear documentation supports review, oversight, and alignment with audit and patient access expectations.
- Support ongoing monitoring after deployment by tracking performance, detecting changes in data or results, and conducting routine reviews. Continuous evaluation helps ensure AI systems remain safe, effective, and fair as clinical practices and patient populations change.
- Support privacy-preserving AI development through the use of synthetic data, enabling model development, testing, and validation when access to PHI is restricted. Synthetic datasets can enable model development and validation without exposing patient identities.
- Support interoperable and traceable integration of AI into clinical workflows, aligned with ONC certification and interoperability requirements. Traceability across systems helps organizations demonstrate appropriate data use, maintain audit trails, and meet information-sharing expectations.
- Support responsible adoption of generative and agentic AI by encouraging governance approaches that prioritize safety, transparency, and setting clinical boundaries, including guardrails when appropriate and enabling human-in-the-loop review for higher impact clinical use cases. These practices help manage risk while allowing innovation to move forward.
- Build workforce readiness through education and training, helping clinicians and staff understand how to interpret AI outputs, recognize limitations, and raise concerns. Improving AI literacy reduces misuse, over-reliance, and loss of trust.
- Encourage consistent benchmarking and evaluation across settings, so organizations can compare model performance and fairness using common data and measures. Shared evaluation approaches reduce uncertainty and help build confidence in AI adoption.

Reimbursement

HHS's payment policies and programs have massive effects on how health care is delivered in the United States, oftentimes with unintended consequences. Hypothetically, if a payer is taking financial risk for the long-term health and health costs of an individual, that payer will have an inherent incentive to promote access to the highest-value interventions for patients. Under government designed and dictated feefor- service regimes, however, coverage and reimbursement decisions are slow. Rarely does covering new innovations reduce net spending; and waste, fraud, and abuse is difficult to prevent, oftentimes leading to massive spending bubbles on concentrated items or services that are not commensurate with the value of such products. Given the inherent flaws in legacy payment systems, we seek to ensure that the potential promises of AI innovations are not diminished through inertia and instead such payment systems are modernized to meet the needs of a changing healthcare system. We seek feedback on payment policy changes that ensure payers have the incentive and ability to promote access to high- value AI clinical interventions, foster competition among clinical care AI tool builders, and accelerate access to and affordability of AI tools for clinical care.

Reform Payment Structures to incentivize AI-enabled Care with transparency and governance to Reflect Value, Not Volume: Traditional fee-for-service (FFS) payment systems, which reimburse based on volume of services, can slow coverage decisions, disincentivize innovation adoption, and inadequately recognize the value of AI-enabled care. Reform aligning financial incentives with health outcomes and efficiency will encourage payers to deploy high-value AI tools and invest in evidence generation.

Policy Changes:

- Create reimbursement pathways that explicitly value AI contributions to care delivery, such as new CPT categories or modifiers for AI-augmented services like clinical note-taking, assistance with diagnosis, and post-operative follow-ups as well as AI-enabled medical devices, so that payment reflects the incremental diagnostic, prognostic, or efficiency benefit of AI.
- Incorporate outcomes-based or performance-driven payments that reward measurable clinical improvements (e.g., accuracy, reduced adverse events, reduced avoidable admissions), aligning reimbursements with the quality and value of AI use rather than procedural counts.
- Expand value-based payment models (VBPMs) that explicitly incorporate AI-enabled care into shared savings, bundled payments, or account-based arrangements, so payers and providers benefit financially but only from improved outcomes.
- Authorize pilot and demonstration programs through entities like the CMS Innovation Center (CMMI) to test AI-focused payment innovations that quantify AI's contribution to cost and quality outcomes.
- Adjust enrollee cost-sharing under value-based insurance design so that high-value AI-enabled interventions are more affordable for patients, increasing uptake and supporting payer adoption.
- Establish transparent reimbursement frameworks that account for the diversity of AI tools (e.g., software as a medical device, integrated clinical decision support, diagnostic assistance) and avoid one-size-fits-all rules.
- Promote standardized evidence and evaluation requirements such that payers and developers operate on common metrics of clinical utility, safety, and cost-effectiveness, lowering market barriers to entry and foster competition for new innovators.

Government reimbursement policy changes—focused on value recognition, outcome-driven payment structures, transparent and flexible pathways, value-based care integration, and evidence generation incentives—can ensure that payers are financially motivated and operationally empowered to promote high-value AI clinical interventions. These changes also help catalyze competition among clinical AI builders and broaden access and affordability of AI tools for patients and providers alike.

Research & Development

HHS supports one of the world's largest health research ecosystems, catalyzing innovation to supplement the market. By enabling applied AI research & development, care delivery research and implementation science, as well as AI entrepreneurship in health care, we can better translate AI technologies from concept to clinical use. We seek input on ways in which HHS may invest in research & development (including public-private partnerships and cooperative research and development agreements (CRADAs)) to integrate AI in care delivery and create new, long-term market opportunities that improve the health and wellbeing of all Americans.

HHS has a unique opportunity to accelerate the translation of AI technologies from concept to clinical impact by building on proven public-private partnership models that have already transformed health research and safety surveillance at scale. Strategic investments in applied AI research, cooperative agreements, and shared infrastructure can both de-risk innovation and create durable market demand for AI-enabled solutions that improve health outcomes for all Americans.

A leading example is the **NIH All of Us Research Program,** where SAS serves as a key industry partner helping enable secure, scalable analytics across one of the most diverse longitudinal health datasets ever assembled. SAS technologies have supported the program's ability to curate, manage, and analyze complex multimodal data (EHRs, genomics, wearables, and patient-reported outcomes) in a manner that is privacy-preserving, auditable, and accessible to a broad research community. This has directly empowered researchers to develop, test, and validate AI models across populations historically underrepresented in clinical research, improving both the scientific rigor and equity of AI-enabled discoveries. The All of Us model illustrates how HHS can stimulate AI innovation not simply through grants, but through platform-level public-private partnerships that provide governed access to high-value data, modern analytic environments, and interoperable tools. By investing further in such collaborative research ecosystems, HHS can ensure that AI solutions are clinically relevant, generalizable, and ready for deployment in real-world care settings.

Similarly, the **FDA Sentinel Initiative** demonstrates how a long-term public-private collaboration can operationalize AI and advanced analytics for direct impact on patient safety and regulatory science. Through Sentinel, SAS has partnered with the FDA, Harvard Pilgrim and a nationwide network of data partners to enable near–real-time safety surveillance of drugs, biologics, and medical devices. This work has advanced sophisticated signal detection, comparative effectiveness, and risk stratification capabilities using real-world data at scale. Sentinel reflects a model where cooperative research and development agreements and shared analytic standards allow AI innovation to mature within mission-critical environments, ensuring that models are not only technically sound but trusted by regulators, clinicians, and manufacturers alike. Expanding this approach into other domains of care delivery would allow HHS to accelerate AI adoption while simultaneously creating long-term market opportunities for validated production-grade AI systems.

Together, these examples suggest that HHS can most effectively catalyze AI in care delivery by investing in durable, interoperable, and privacy-preserving research infrastructure, co-developed with industry and academia, rather than isolated pilot projects. By scaling or creating new partnership-driven platforms like All of Us and Sentinel, HHS can ensure that AI innovation is not only rapid, but clinically credible, economically sustainable, and broadly accessible across the U.S. healthcare system.

Specific Questions

In addition to the general requests for information above regarding AI regulation, reimbursement, and research & development, HHS seeks input on the following specific questions:

1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

Despite meaningful progress in digital health transformation, persistent data fragmentation and interoperability challenges continue to limit the speed, scalability, and reliability of AI innovation. Compounding these structural barriers, many health care professionals lack sufficient education in analytics and AI, which undermines confidence and trust in AI-driven solutions. This trust gap is further exacerbated by limited transparency into how models are developed, governed, and validated, and their unclear or unproven impact to clinical outcomes.

Structural Data Barriers:

Data fragmentation. The availability of new sources of health and social determinant data is expanding rapidly with the proliferation of wearables, remote monitoring devices, genomics, digital biomarkers, and patient-generated health data. Patients are offered options to have continuous visibility into their health, and clinicians increasingly recognize the value of real-time signals such as sleep quality, heart rate variability, and other vital signs. This growth in data sources has, however, intensified fragmentation. Critical information remains distributed across disparate systems, vendor platforms, and emerging point solutions, making it difficult for robust, unbiased AI models to be developed and trained because patient records are incomplete.

Unstructured Data. In addition, a significant portion of healthcare's most valuable information lives in unstructured formats, including free text clinical notes, PDFs, scanned documents, images, and even faxes. These sources often contain contextual nuances that drive accurate clinical interpretation and high performing AI models. While progress is being made through technologies such as **SAS Document Analysis for Health Records** (SAS Models | SAS), natural language processing, and large language models (LLMs), the extraction and normalization of this unstructured content is still a major operational challenge which contributes to fragmentation.

Interoperability and Workflow Challenges. Even with the industrywide commitment to standards such as Fast Healthcare Interoperability Resources (FHIR), interoperability remains inconsistent and unreliable. Data exchange varies significantly across electronic health records (EHRs), health information networks, and digital health tools, often requiring costly bespoke integrations or manual workflows. These inconsistencies hinder the ability to build scalable AI solutions that integrate into existing workflows. Providers will be unwilling to adopt if there are additional steps or actions which require them to work outside of their source system or existing workflows.

Without the ability to access de-identified, accurate, and comprehensive whole patient records, innovators are unable to validate and test the scalability of AI-enabled innovations.

SAS can help overcome these barriers through health data platform solutions to empower the care continuum with deep analytics capabilities and insights. These solutions, including SAS Health, automate the integration and processing of vast volumes of data from multiple sources, allowing healthcare organizations to make informed decisions across their workflows while providing secure data ingestion and management, adhering to FHIR specification standards to facilitate healthcare data interoperability, and supporting the Outcomes Partnership Common Data Model standard to visualize and analyze observational data. These solutions facilitate the design and execution of analysis on industry-standardized clinical data. The inclusion of embedded AI/ML enables predictive insights, supporting proactive decision-making across healthcare organizations, catering to various healthcare use cases, from population health management to financial oversight.

As a part of our platform solution, we offer SAS Document Analysis for Health Records which unlocks the rich clinical detail buried in unstructured notes by transforming scanned PDFs, faxed records, and narrative documentation into structured, analytically ready data. Using advanced Optical Character Recognition and clinically informed, deterministic extraction models, data can be extracted to round out otherwise lacking data sets. This automated workflow increases efficiency, reliability, and auditability while overcoming common challenges such as dirty data, repetitive progress notes, and multi-hundred-page record sets, making once-inaccessible insights available for downstream analytics and decision support.

The deterministic capabilities of document analysis are complemented by a contextual intelligence layer that can retrieve, assemble, and present the specific evidence, policies, guidelines, and historical precedent required to support the data, or furthermore, support the analytic result.

Workforce Enablement: To fully realize the potential of analytics and AI in healthcare, targeted education and workforce enablement programs are urgently needed, similar in scale and intent to the Meaningful Use retraining initiatives that accompanied the adoption of electronic health records. Just as federal investment and structured education were critical to helping clinicians effectively implement and trust electronic health record (EHR) systems, comparable funding and coordinated training are now required to build analytics and AI literacy. Healthcare professionals need to be educated on how to interpret AI outputs, fostering the collaboration of human experts and AI systems. Without deliberate investment in education, change management, and applied learning, AI innovation risks outpacing the workforce’s ability to adopt, govern, and use it safely and effectively.

As the role of the health care professional is fundamentally evolving as AI becomes more deeply embedded in clinical and operational decision-making, rather than serving as continuous manual operators, clinicians and allied health professionals will be increasingly positioned as safety control engineers, overseeing, validating, and intervening in AI-driven processes when anomalies, uncertainty, or risk emerge. This evolution closely mirrors the transformation of pilots in modern aviation, whose responsibilities have shifted from direct control to supervisory oversight of highly autonomous systems, intervening primarily in exceptional or emergency scenarios. To ensure this transition improves care quality and safety rather than introducing new risks, rigorous, interdisciplinary research is needed to define these emerging responsibilities and to deliberately redesign the roles, workflows, and training of both direct care providers and their support teams in an AI-enabled health care system.

Addressing the Trust Gap: SAS has long demonstrated a sustained commitment to responsible and trustworthy AI by embedding governance, transparency, and accountability across the full analytics and AI lifecycle. As AI adoption accelerates and regulatory expectations evolve, SAS has consistently emphasized the importance of balancing rapid innovation with rigorous oversight to ensure AI systems are reliable, ethical, and aligned with societal and organizational values.

Operationally, SAS advances trustworthy AI through an integrated governance framework built on its platform, enabling organizations to manage, document, validate, and audit AI models throughout their entire lifecycle. These AI governance capabilities provide centralized visibility into AI use cases, models, and agents, while embedding policy controls, risk management, and human-in-the-loop oversight at critical decision points. By treating AI governance not as a standalone function but as the next stage of maturity in enterprise governance, SAS supports customers in scaling AI responsibly across industries, including healthcare, where transparency, fairness, and regulatory readiness are essential.

At the platform level, we deliver practical, clinician- and stakeholder-friendly transparency through capabilities such as model cards that function as “nutrition-label-like” summaries for AI models. These model cards provide clear, easy-to-read insights into a model’s intended use, training data, performance, fairness, and limitations, helping healthcare professionals and decision-makers quickly understand what a model does, how it was developed, and where it can be trusted. By embedding these trustworthy AI features directly into the analytics workflow, SAS reduces friction between innovation and governance, enabling healthcare organizations and its front-line workers to adopt AI with greater confidence, clarity, and measurable value.

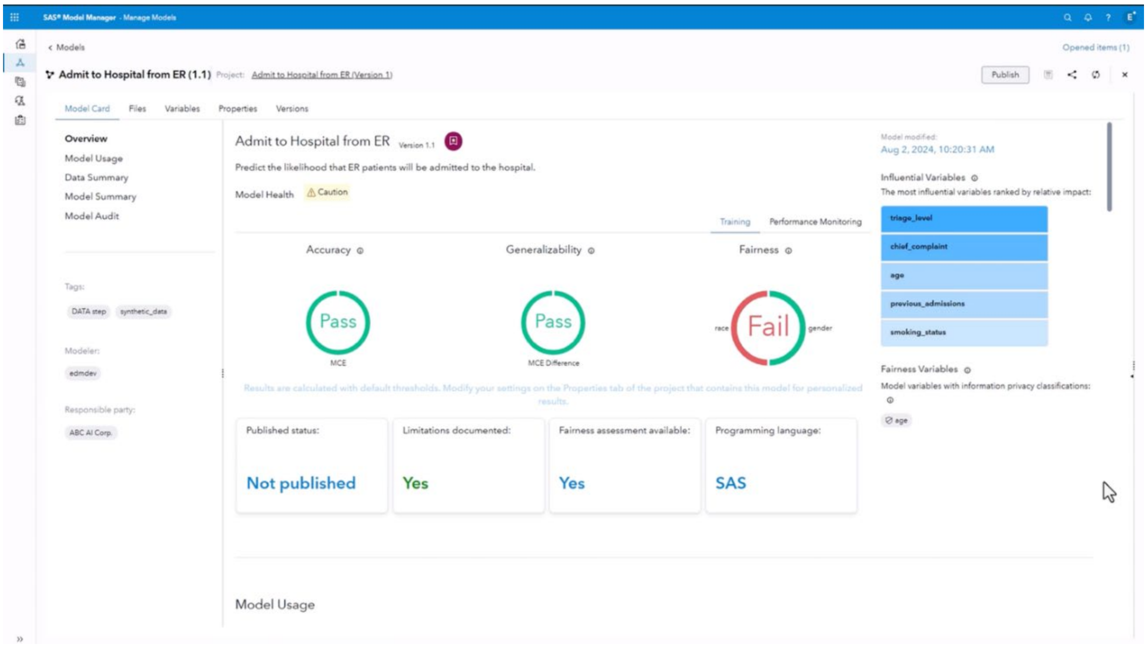

Figure 1. SAS AI Model Cards for Analytics Workflow.

2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

N/A

3. For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (*e.g.,* relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?

As AI is increasingly embedded into clinical care through applications that fall outside traditional medical device classifications, new legal and implementation challenges emerge that are not fully addressed by existing governance and accountability structures. These challenges are less about individual algorithms in isolation and more about how AI systems are governed, monitored, and held accountable across their lifecycle within complex healthcare environments.

A central issue is accountability in hybrid human-AI decision-making. In many non-medical device use cases, AI informs but does not directly execute decisions, creating ambiguity around responsibility in adverse outcomes. Questions remain regarding how liability should be shared among developers, healthcare organizations, and clinical users, and how traceability should support legal review and indemnification. SAS's experience across regulated industries demonstrates the importance of clear governance structures, auditable documentation, and defined roles spanning development, deployment, and monitoring of AI models. HHS can play a constructive role by promoting consistent accountability and documentation standards for AI that is used in clinical contexts.

Privacy and data protection present another major challenge, particularly as AI relies on large, diverse datasets for training, validation, and continuous learning. Risks related to re-identification, secondary use, and expanding exposure surfaces grow as models evolve. SAS strongly supports privacy-by-design approaches that embed anonymization, access control, and data minimization into AI architectures from inception. In this context, synthetic data offers another powerful mechanism to enable innovation while reducing exposure to sensitive patient information. In higher-risk scenarios, federated deployment approaches, where models are trained locally and production data never moves from its original location, provide a viable path to minimizing centralized data risk and government data access laws. HHS can accelerate safe adoption by recognizing synthetic data and federated models as acceptable, privacy-preserving methods for clinical AI development and validation.

Transparency and explainability are equally essential for legal defensibility and clinical trust. AI systems often lack formal premarket review, making documentation, performance clarity, and model traceability critical. SAS's AI governance approach emphasizes and automates rigorous documentation of assumptions, risks, bias testing, and mitigation strategies, alongside version control and decision traceability. HHS can strengthen its capabilities in this area by creating policy around minimum expectations for AI explainability and transparency that are aligned with federal AI trust and civil rights principles.

Bias, fairness, and equity risks are particularly acute for AI used in triage prioritization, care access, and resource allocation. Without formal oversight, biased outcomes may persist undetected, exposing providers to legal and reputational risk. SAS's experience supporting audit and compliance in financial services demonstrates that continuous bias detection and routine audits must be in the operational requirements of any AI used in clinical care. HHS can advance responsible AI by encouraging standardized bias and fairness assessments for AI deployed in federally supported care environments.

Finally, most healthcare organizations lack scalable governance infrastructure to manage AI across departments and use cases, leading to fragmented oversight and elevated compliance risk. SAS's pragmatic AI framework emphasizes continuous risk management across the AI lifecycle, informed by decades of experience in highly regulated industries where transparent, scalable governance is foundational. This experience now directly informs a generalizable AI Governance approach for healthcare, enabling organizations to inventory AI systems, monitor performance and drift, audit decisions, and enforce policy consistently.

HHS has a critical role not only as a regulator but as a market shaper. We encourage HHS to promote AI governance as core infrastructure rather than an optional compliance standard, supporting reference architectures for responsible AI, and embed governance expectations into funding, reimbursement, and accreditation programs. This would allow HHS to accelerate adoption of AI that is not only innovative, but legally defensible, operationally resilient, and worthy of public trust.

4. For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (*e.g.,* contracts, grants, cooperative agreements, and/or prize competitions)?

Building on the governance and accountability challenges described above, effective evaluation methods are how those principles are put into practice in clinical settings. As AI is increasingly used in clinical care outside of traditional medical device categories, strong evaluation methods are essential to ensure safety, effectiveness, fairness, and trust over time. Based on SAS' experience supporting analytics and AI across highly regulated industries, including healthcare, these evaluation practices must recognize that many AI systems continue to change after deployment. As a result, evaluation cannot be a one-time activity and must occur both before and after deployment in real clinical environments.

Before deployment, evaluation should focus on whether an AI system is appropriate for its intended clinical purpose. Traditional performance measures like accuracy, sensitivity, and specificity remain important, but they are not enough on their own. SAS' AI governance approach operationalizes pre-deployment evaluation by embedding performance, bias, and robustness testing directly into the model development lifecycle. Drawing on SAS' approach to responsible AI, systems should also be evaluated for how well they perform across different patient populations, how they handle incomplete data, and how consistently they perform across care settings. Scenario-based testing, including the use of synthetic data, can help identify weaknesses before an AI system is used in patient care. Bias and fairness testing should also occur upfront, with results clearly documented and addressed when disparities are identified. SAS' governance framework emphasizes documentation of these findings through standardized model documentation that captures intended use, limitations, and evaluation results. Evaluation should also account for how AI fits into real clinical workflows. Simulation testing, clinician-in-the-loop review, and usability testing helps to assess how AI outputs are understood, trusted, and acted on in practice. From SAS' experience, these methods can reveal risks that may not appear in technical testing alone. Clearly documenting intended use, limitations, and when human judgment should trump AI recommendations could lead to safer implementation and clearer accountability.

After deployment, continuous monitoring becomes critical. Many current AI governance approaches fall short in this area. SAS's experience in operationalizing AI at scale highlights the importance of ongoing evaluation to track model performance over time, detect data drift, and assess whether changes in patient populations or clinical practice affect outcomes. Useful post-deployment measures include how often AI outputs align with clinical judgment, how frequently recommendations are overridden, how stable the predictions are, and whether the impacts differ across demographic groups. Routine audits, version tracking, and assessments of model updates also help to ensure that changes do not introduce new risks. Also, traceability (i.e., linking predictions to specific model versions, inputs, and contexts) is essential for accountability and retrospective review.

Additionally, post-deployment evaluation should examine how AI is actually used by people in practice. Even systems that perform well can create risks if clinicians don't understand how to interpret the outputs or recognize limitations. Consistent with SAS's approach to analytics, evaluation should also include whether users understand when AI can appropriately support decisions, when caution is warranted, and how concerns can be raised or escalated. Training and feedback methods that build basic AI literacy and reinforce appropriate use can help reduce misuse, over-reliance of AI, and loss of trust.

SAS believes that HHS can meaningfully support these evaluation efforts by investing in both the infrastructure and the people. Grants and cooperative agreements could support the development of standardized evaluation frameworks, common workflows, and benchmarking approaches, as well as fund education and upskilling for frontline healthcare workers in areas such as AI literacy and analytics. Contracts could be used to pilot ongoing monitoring and evaluation approaches in federally supported healthcare settings. In addition, prize competitions, challenge programs, and healthcare-focused hackathons can encourage innovation in evaluation methods while bringing together clinicians, data scientists, and patients to solve real-world problems. Based on SAS' experience supporting and participating in large-scale analytics and AI hackathons, these forums can be particularly effective for testing practical use cases, advancing evaluation approaches, and accelerating shared learning across the healthcare ecosystem. For example, the [2025 SAS Hackathon Champion Change Bringers] team developed an AI-powered monitoring solution for Parkinson's disease symptoms, illustrating how collaborative, problem-focused competitions can advance evaluation methods, human-centered design, and practical workflows in clinical AI.

5. How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

Hospital Lillebælt Denmark modernized emergency-department diagnostics through the integration of AI, automation, and real-time data for purposes of improving traditional emergency-care workflows which rely heavily on laboratory diagnostics and can often result in delayed triage and treatment during critical moments.

SAS applauds the administration's focus on how AI can be leveraged by clinicians to better care for patients. SAS agrees that the AI revolution has brought and will continue to bring innovation and value to health care and specifically clinical care. SAS is used by hundreds of health care providers across the world, and we've had incredible success in reducing clinician burden, providing actionable clarity from health data and predicting negative outcomes before they happen. Through this work, we also recognize that not all clinicians are data scientists and that evidenced based healthcare may not always understand the potential for bias as risk with AI. This is why SAS has developed what we describe as model cards[1][2]. SAS proposes that our platform (specifically our capabilities associated with registering and evaluating

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[1] https://blogs.sas.com/content/subconsciousmusings/2024/08/08/sas-model-cards-now-available/

[2] https://video.sas.com/detail/video/6360057183112/model-card:-nutrition-label-for-an-ai-model

models) serve as the foundation technology for a proposed, governed assurance lab. Innovators can register AI/ML models (developed in SAS or open source) into a governed repository. This ensures that all models entering an assurance workflow are versioned, traceable, and auditable. SAS can automatically generate model cards (sometimes referred to as scorecards), which summarize a model's metadata, training data, performance metrics, fairness indicators, assumptions, and limitations.

6. Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?

SAS can provide numerous case studies of where AI tools have been deployed in clinical care. Following are four case studies which exemplify where AI tools have improved outcomes or addressed the rising cost of care.

Case Study #1: Taipei Veterans General Hospital

Taipei Veterans General Hospital, one of Taiwan's leading medical centers, sought to improve outcomes for the country's 90,000+ hemodialysis patients, many of whom face complications such as heart failure, anemia, and fluid-management discomfort.

Using **SAS AI and real-time IoMT streaming capabilities**, the hospital implemented models which were designed to identify patients in real time, with heart-failure risk. Furthermore, models assisted clinicians in their efforts to optimize dry-weight targets and detect anemia.

The results were impressive and reports noted 90% accuracy in heart-failure risk prediction and an 80% reduction in dry-weight adjustment errors. Healthcare professionals were able to identify anemia earlier and reported that by having comprehensive dashboards impacted the amount of time to round on the dialysis unit. They reported a reduction in rounding time from 20 minutes (and up to 40 minutes), reduced to 3 minutes.

By integrating streaming data from dialysis machines, national health-insurance databases, health records, and chest-x-ray imaging, clinicians now receive real-time alerts and transparent, interpretable model insights that support clinical decision-making.

Agentic AI could further enhance the current workflow which autonomously monitors streaming IoMT data, dynamically invoking predictive and analytic models, by coordinating follow-up actions such as alert prioritization and workflow routing.

Case Study #2: Hospital Lillebælt

Hospital Lillebælt in Denmark modernized its emergency-department diagnostic workflow by integrating AI, automation, and real-time data, transforming traditionally lab-dependent processes that often delay triage and critical interventions. This advanced approach has streamlined diagnostic throughput, and reduced time-to-decision, enhancing the clinicians' ability to deliver timely, effective emergency care.
In this implementation, the flow of real-time laboratory, clinical, and operational data and automated data ingestion and event-stream processing are orchestrated in real time, trigger AI-driven diagnostic models. Clinicians receive both diagnostic interpretations and clinical decision-support recommendations within 60 minutes, enabling faster triage, earlier identification of high-risk patients, and more streamlined coordination between emergency teams and laboratory services.

This end-to-end automation ensures a consistent, repeatable, and high-quality diagnostic workflow that improves patient outcomes while reducing clinician burden.

Agentic AI could further enhance this workflow, by autonomously coordinating downstream operational actions, such as proactively reserving an appropriate hospital bed, when critical clinical insights are surfaced. As an example, when high-risk signals are detected, agentic AI can, continuously monitor patient acuity signals across clinical, laboratory, and IoMT data streams, Interpret predictive risk outputs in operational context (such as current bed availability, unit capabilities (ICU vs. step-down), and staffing constraints), initiate pre-emptive bed reservation workflows, signaling bed management and patient flow systems to hold or prioritize an appropriate bed and coordinate cross-team actions by notifying emergency, inpatient, laboratory, and environmental services teams in parallel.

Case Study #3: Large Healthcare Payer

A healthcare payer wanted to understand the benefits of addressing members with behavioral and medical comorbidities. Using SAS, they examined their claims data to better understand which members presented with behavioral vs. medical conditions, which members presented with both and what the impact was after proactively identifying members with behavioral conditions and engaging them with their behavioral health product.

The analysis revealed that 82% of adult claimants who presented with behavioral conditions also have medical conditions and 27% of adult claimants who presented with medical conditions also have behavioral conditions.

Furthermore, the analysis also revealed that there was a 329% increase in behavioral health claims during 2020 to 2023, resulting in a 110% increase in behavioral health spending, and 25% of members with behavioral health conditions drove 43% of the book of business spend. The costs for those conditions were noted to be 3-6 times higher for members who had not received behavioral health treatment.

Predictive analytics were built to proactively identify members for outreach, for purposes of directing them to programs offered by the payer's existing flagship behavioral health product. As a result of this initiative:

- $193 per member per year was experienced in medical cost savings
- $2,505 in cost savings per member in the 15 months post-diagnosis of a behavioral condition
- There was a 27% increase in utilization of their Behavioral Health Flagship Product, including an increase in virtual behavioral health claims
- 15% more members received care

Case Study #4: The Sax Institute

Headquartered in Sydney, Australia, the Sax Institute is not-for-profit organization that brings research to the frontlines of policy, programs and services. They connect more than 70 public health research organizations across Australia into a national network through the **Secure Unified Research Environment (SURE) platform, powered by SAS.**

More than 800 researchers and 300 SAS users rely on SURE to access nationally linked health data. Over **366 active research projects** have been conducted using SURE, with the Institute's most ambitious initiatives being:

- 45 and Up Study - Australia's largest ongoing investigation into **health and aging**
- 18 and Up Study - intergenerational study on **mental health**
- **Digital twins** to simulate hospital workflows and model system-wide changes to improve patient care

7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

Frontline workers. AI adoption in clinical care is most successful when it is driven by the people closest to patient care, frontline workers. While executives, clinical leaders, and governance committees play essential roles in prioritization and oversight, nurses, physicians, allied health professionals, technicians, and administrative support staff exert the greatest real-world influence on whether an AI solution is trusted, utilized, and integrated into daily workflows. Their lived experience, tacit knowledge, and understanding of operational realities determine whether an innovation becomes an asset or an added burden. Frontline workers can identify areas where AI can have the highest impact on patient outcome, efficiency and clinician support.

Because of this, frontline teams must be deeply involved in the design, testing, and iteration of any AI-enabled workflow. Strict adherence to design thinking principles (empathy, co-creation, rapid prototyping, and iterative feedback) is critical to ensure solutions align with actual clinical needs and reduce friction at the point of care. When frontline staff are co-designers rather than recipients of change, adoption accelerates, trust increases, and AI becomes an enabler rather than a disruptor.

In parallel, organizations should establish and socialize a qualitative and quantitative return on investment (ROI) measurement framework during the design phase. This tool should define expected benefits (e.g., time saved, reduction in cognitive load, improved throughput, enhanced patient safety, improved documentation quality), and outline how those benefits will be captured, monitored, and reported. Measuring actual value in a structured, transparent way enables health systems to demonstrate evidence of impact, justify scaling, and build internal confidence. Clear evidence of ROI is often the catalyst for exponential, organization-wide adoption of successful AI innovations.

From an administrative standpoint, there are several hurdles which can be overcome by implementing an agreed upon governance structure, where cross-functional teams make decisions about data, clinical workflows, and technology. They can ensure alignment on an ROI measurement approach, define accountability for AI safety, validation, and continuous monitoring and procurement and compliance cycles that are not designed for iterative or rapidly evolving AI technologies.

SAS, in our continued partnership with Duke University, has developed a whitepaper which outlines a pragmatic AI framework for health care providers. The whitepaper can be accessed via this **link.**

8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

Enhanced interoperability would most accelerate AI for clinical care in areas where it could improve accessing, harmonizing, and validating high-value clinical data across settings. Improving consistency and portability for core data types like longitudinal electronic health record (EHR) data (e.g., diagnoses, encounters, medications, vitals), lab results, imaging metadata, and unstructured clinical notes, paired with standardized representations like HL7

Fast Healthcare Interoperability Resources (FHIR), may offer the greatest opportunity. These data types are foundational for high-impact AI use cases including care management and clinical decision support but remain difficult to use due to inconsistencies across organizations.

In addition, greater adoption of open file formats (e.g., csv, json, parquet) and standards-based APIs could expand market participation by enabling more organizations to securely exchange data and operationalize AI without custom integrations. These open formats support scalable analytics pipelines, reproducible benchmarking, and more seamless collaboration across platforms. Additionally, APIs can enable near real-time data exchange necessary for timely clinical decision support.

App-level interoperability is also a growing priority but is still secondary to core system-to-system exchange. Until foundational interoperability across EHRs, labs, and pharmacies is consistent and reliable, the value of app integration will be limited. However, as the infrastructure matures, app-level access becomes essential for patient-facing innovation and real-time decision support, particularly when paired with governance and auditability that support responsible AI adoption.

SAS and our healthcare partners have already demonstrated the value of enhanced data exchange and the harmonization for advancing clinical research and innovation. For example, through our collaboration with **AstraZeneca**, SAS helped redesign clinical and patient data flows to enable greater automation, interoperability, and the flexible use of diverse patient data sources in support of research and regulatory reporting. This in turn accelerated insights from clinical data and supported more efficient decision making across teams. SAS stands ready to partner with HHS and the broader HHS ecosystem to advance interoperability that widens market opportunities, promote research, and will accelerate the development and safe use of AI in clinical care.

9. What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

Implementation science research has consistently shown that caregivers find themselves excluded from participating in a patient's healthcare team (doctors, nurses, etc.). For example, within the Veterans Affairs Administration, family caregivers have felt under-utilized as part of the care process[3]. AI-driven tools could be used to better incorporate caregiver feedback, as well as provide training to practitioners to shift the culture of care toward improved incorporation of family caregivers.

A key part of this process would be developing automated tools for network analysis and predicting acute patient needs, both of which are enhanced through the use of AI-driven tools, backed by expertise from case managers and physicians.

SAS has proven such capabilities through our Model Manager tool and other offerings, which would allow stakeholders to continually evaluate patient outcomes relative to caregiver interactions and feedback.

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[3] Training Health Care Practitioners to Include Family Caregivers with Web-Based Learning Modules: https://www.thepermanentejournal.org/doi/pdf/10.7812/TPP/22.037?download=true&__cf_chl_tk=E..MKiubJ3Dk4CrX0hZTV5Y.iBVHfwW3cOi2a9u3UNM-1770404188-1.0.1.1-zo0.Kl3fbancmlzPyyx4_cO3V_wcY6t3zRvoSyTe.u0

10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

As has been discussed at length, the impact of AI in clinical care is often slowed with resulting output being biases by incomplete or less than optimal data due to the appropriate controls (including HIPAA, IRB, Declaration of Helsinki, etc.). **SAS recommends that HHS prioritize the evaluation and use of synthetic data built using AI techniques that create representative synthetic datasets from existing real clinical data.** This would vastly increase the amount and easy of use of clinical synthetic data and allow for the quick advancement of AI.

In addition to the evaluation and use of synthetic data; HHS should explore the impact of AI on workforce. When HHS implemented the Meaningful Use program, one of the most consequential and often underrecognized elements was the federal investment in workforce retraining to support nationwide EHR adoption. While the program varied across states and institutions, several patterns emerged that are instructive for today's AI landscape. The initiative demonstrated that **technology adoption at scale cannot succeed without parallel investment in human capability**. Workforce shortages and skill gaps remain among the most pressing constraints in healthcare today. Although Meaningful Use training focused primarily on system navigation and compliance tasks, it was not built to prepare clinicians for advanced data-driven workflows, cognitive load shifts, algorithmic decision support, or continuous cycles of technological change. Research is urgently needed to understand how AI reshapes clinical workflow, including the redistribution of cognitive tasks between clinicians and AI agents, shifts in clinical judgment processes when interacting with algorithmic recommendations and identification of new competencies.

a. Are there published findings about the impact of adopted AI tools and their use clinical care?

There is extensive and growing research demonstrating the impact of artificial intelligence on healthcare quality, safety, and operational performance. Across peer-reviewed literature, real-world deployments, and regulatory-cleared solutions, AI has been shown to improve early detection of patient deterioration, reduce preventable adverse events, lower mortality, and support more efficient care delivery. Importantly, the strongest evidence comes from AI solutions that are embedded directly into clinical workflows, leverage routinely collected electronic health record (EHR) data, and are designed to augment, not replace, clinical judgment. These systems consistently demonstrate that timely, explainable insights enable clinicians to intervene earlier, prioritize care more effectively, and improve patient outcomes at scale.

A well-documented example is the Rothman Index, an AI-driven, real-time illness severity index developed using SAS analytics and widely deployed in hospital settings. The Rothman Index synthesizes nursing assessments, vital signs, and laboratory results into a single, continuously updated score that reflects how sick a patient is and whether their condition is improving or deteriorating.

Under controlled conditions, hospitals using the Rothman Index observed a statistically significant (30%) reduction in risk-adjusted mortality, with observed mortality rates materially lower than. In addition, **deployments of the Rothman Index have been associated with fewer unplanned transfers to the ICU, reductions in emergency escalations such as code blue events, and improved clinician efficien**cy by allowing care teams to focus first on the sickest patients. These outcomes underscore how clinically grounded, explainable AI, when rigorously developed, validated, and embedded into workflow, can deliver tangible improvements in patient safety and quality of care.

Linked is a full list of peer reviewed articles and abstracts.

b. How does the literature approach the costs, benefits, and transfers of using AI as part of clinical care?

For areas beyond accuracy and clinical relevance, literature has focused extensively on time savings and efficiencies gained due to AI in the clinical space but rarely goes beyond to explain what the impact of these efficiencies will be. SAS suggests that HHS in its evaluation of AI in the clinical setting examine the impact beyond accuracy and efficiency for a single task but look at the larger impact to the complex system that is healthcare. While it's unclear what AI in healthcare will ultimately bring, there are examples from the past that HHS can look to in identifying potential unintended consequences.

The expansion of electronic health records (EHRs) moved forward by the HITECH Act of 2009 has successfully electrified most health records in the United States but continues to fall short on the goal of true interoperability. In addition, the use of EHRs has led to more optimized billing by providers, leading to an increase in expenditure without the associated efficiencies promised. HHS should ask not what efficiencies are gained by AI, but rather who those efficiencies will be utilized by in multi-trillion-dollar system that is healthcare. The goal will be to increase the quality and availability of care and the same, if not lower cost, but the risk of overuse or continued burn out as clinicians are asked to do more and more, while being moved away from the patient remains a risk.

Appendix: SAS Health

SAS Health - Business Benefits

Reduce Time to Analytics Value: Easily ingest health care data from standard industry formats.

Cross-functional Collaboration: More streamlined work processes among different teams.

Reduce Duplication And Complexity: Standardized data model for confident decisions and speed to value.

Reporting And Insight Generation: NLP & AI to empower clinical/business teams to translate raw data into clear insights, guidance and out-of-the-box reports to identify trends and outliers

Seamless Integration and User Experience: Deploy sophisticated analytics across the entire organization.

Out-of-the Box Applications: Solutions for business challenges, with minimal customization & strong support.

SAS Health

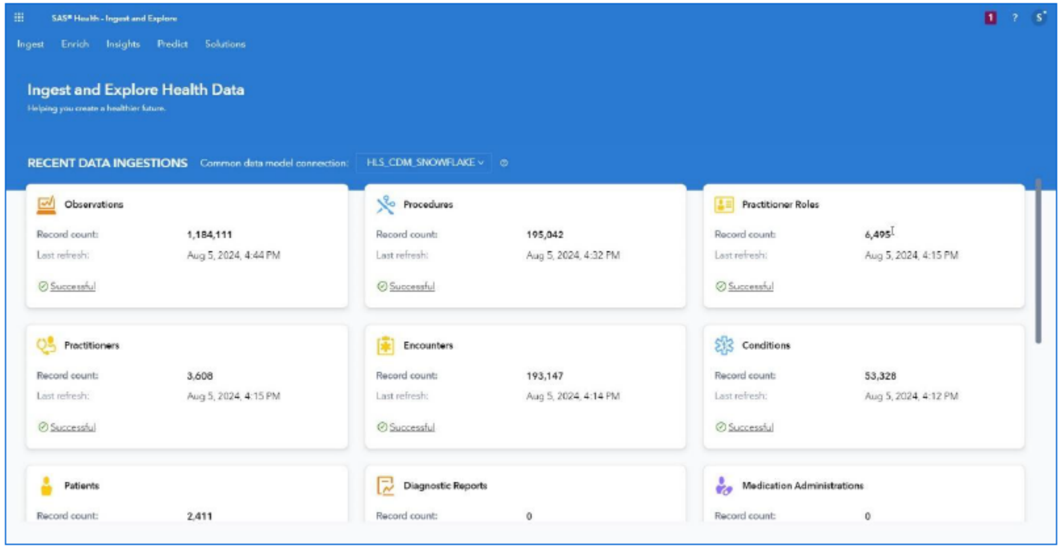
End-to-end solution for health data integration, management & analytics

MAIN FEATURES

Simplifies health data management and accelerates analytic discovery

Data adapters for industry-standard automated ingestion

Common health care data model built for analytics



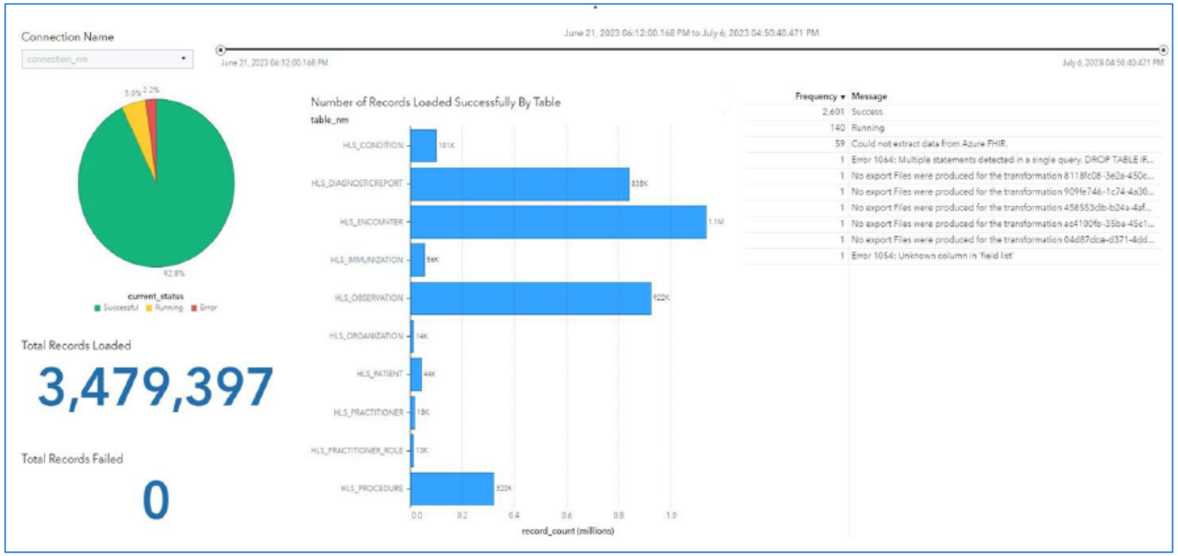
SAS Health

Secure data access & regulatory compliance

MAIN FEATURES

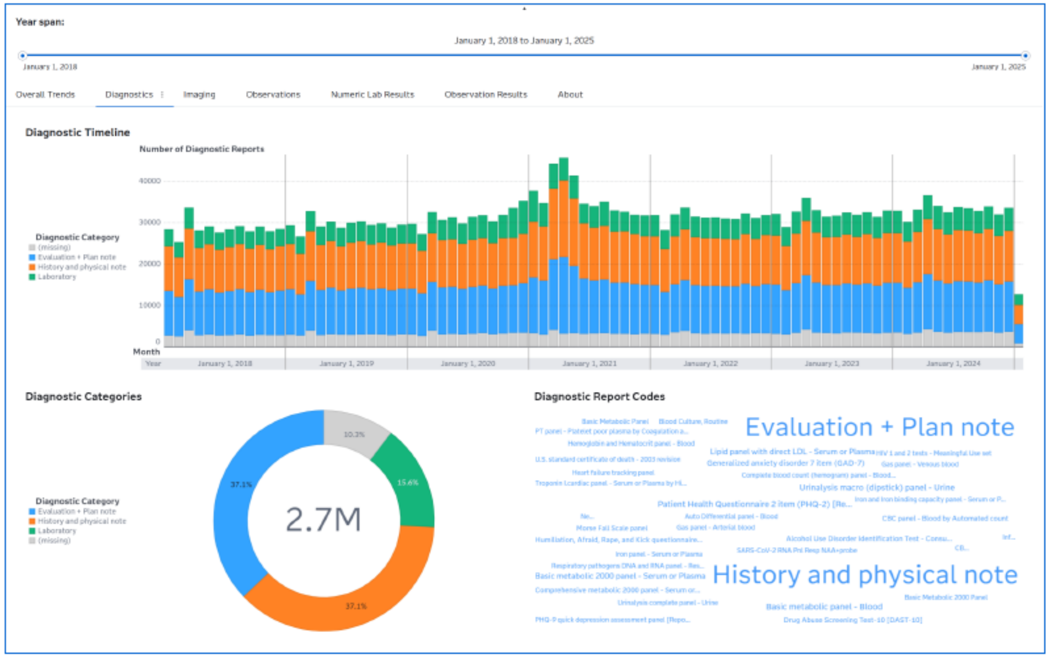
Establish and manage secure access to health care industry specific data sources, applications and systems

Schedule routine data ingestions at your convenience



SAS Health

Insights you can trust



MAIN FEATURES

Foster trust and independence by empowering all users to explore data

Low-code, no-code interface

From Data to Insights- all seamlessly in one place

Ingesting data into CDM*

Creating Patient groups

Generating reports Building models

Analyzing the results

Making data-driven decisions Taking action

SAS® Health on SAS® Viya

*CDM=SAS Health Common Data Model

SAS Health

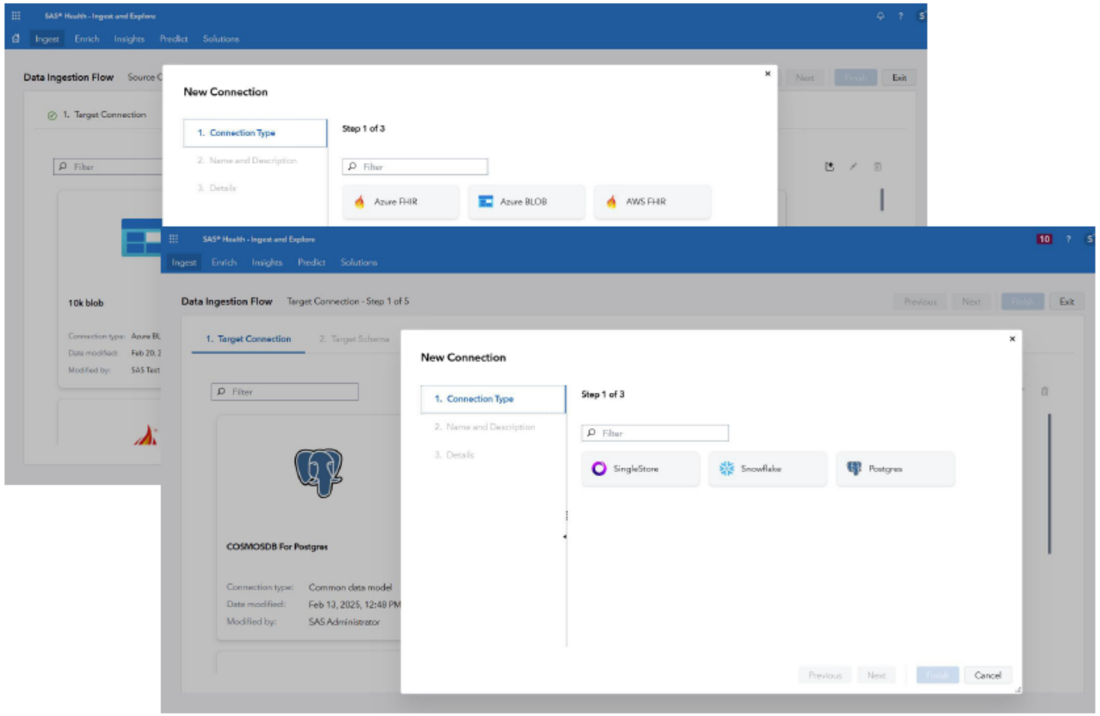
Ingest data into a standardized health care data model

MAIN FEATURES

Easily ingest data from FHIR servers, FHIR Bundles or via CSV into a health care data model

Effortlessly transform data into a standardized healthcare data model

Smooth integration and storage in leading platforms such as SpeedyStore (SingleStore), Snowflake or PostgreSQL.



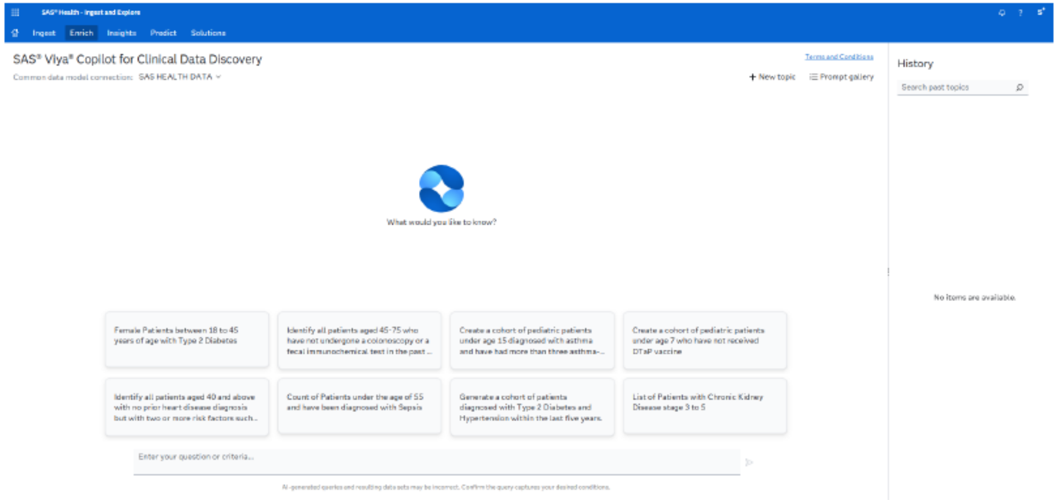
SAS Health

SAS Viya Coplot for Clinical Data Discovery

MAIN FEATURE

Translate raw data into clear insights and guidance without needing technical expertise, helping health care professionals create reports and make informed decisions about patient care, operations and costs.

Use transparent AI/ML predictions to enhance descriptive analytics and improve decision-making.



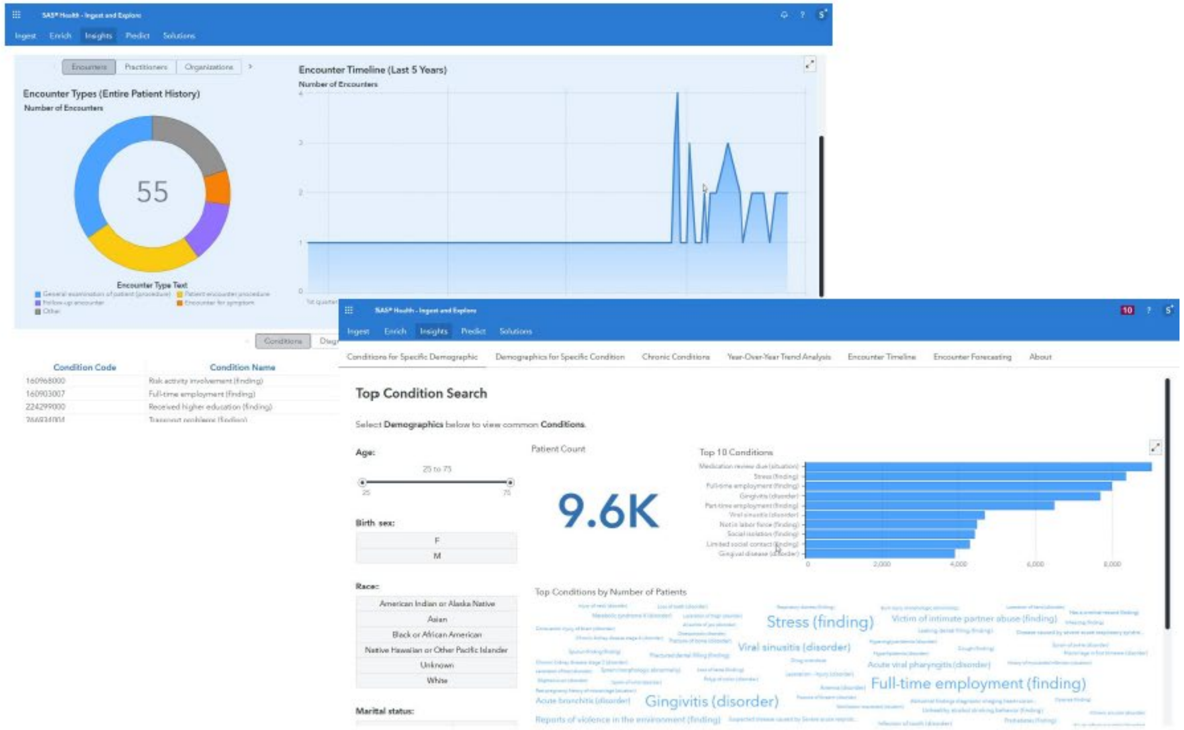
SAS Health

Generate reports and build models

MAIN FEATURES

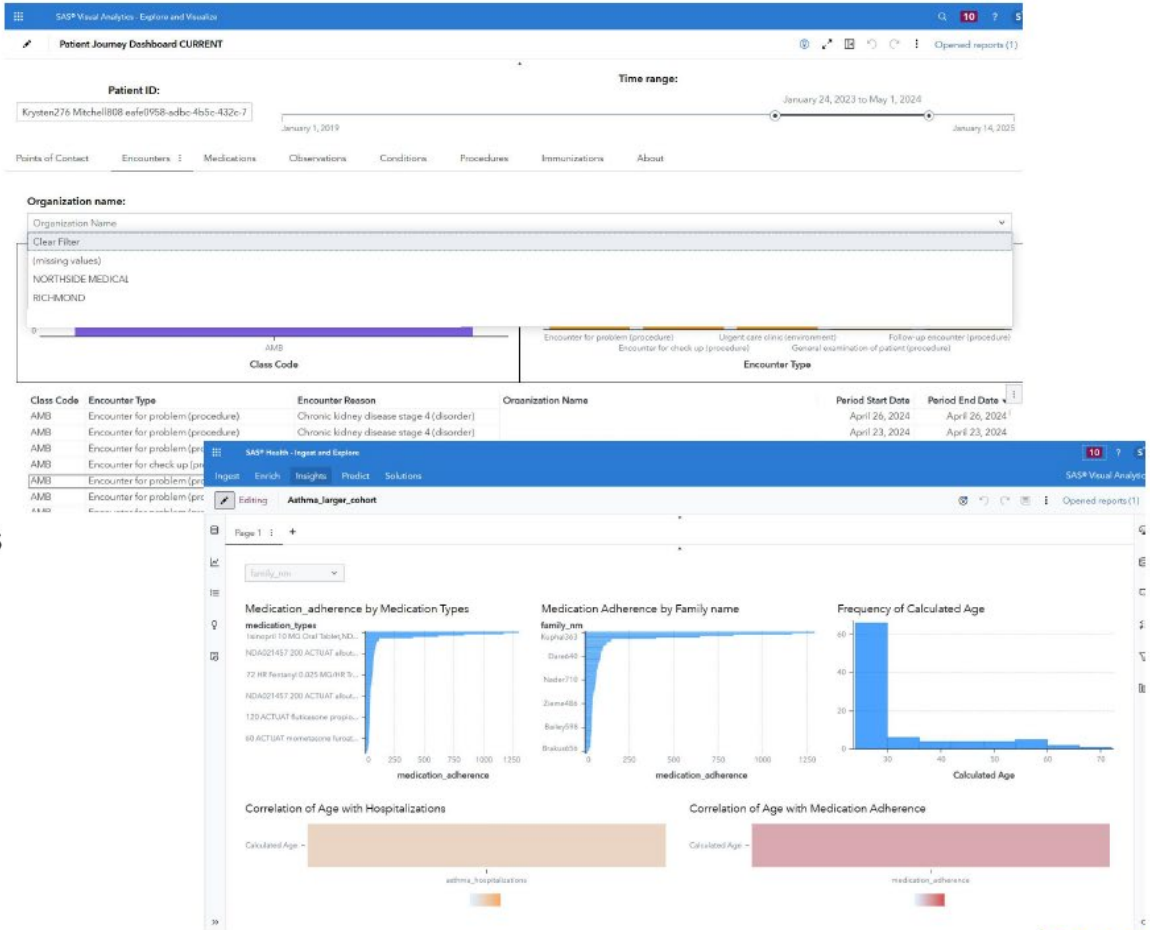
Unlock the power of SAS Viya to improve decision-making and reduce complexity

Business and clinical users benefit from a low-code/no-code interface



SAS Health

Analyze the results



MAIN FEATURES

Analyze results for actionable insights, driving informed decisions and improved outcomes

Use new insights to identify trends and outliers

Interactive Response Index

Thank you

We see great potential in what we can accomplish together.